Transmission Of Phytoplasma Through Vectors

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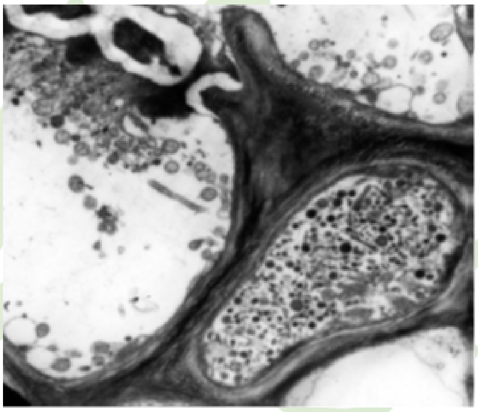
Introduction

The phytoplasmas are phloem inhabiting bacteria, unicellular parasitic prokaryotes. The mechanisms of dissemination are mainly by the vector insects. Phytoplasmas have sizes variable from 200 to 800 nm, they are polymorphic because of the lack of cell wall and multiply in the isotonic environments provided by plant phloem and insect hemolymph.

The common vectors are members of the order Hemiptera, from the families Cicadellidae, Cixiidae, Psyllidae, Cercopidae, Delphacidae, Derbidae, Menoplidae and Flatidae.They were discovered in 1967 by Japanese scientists in mulberry plants who termed them as Mycoplasma like organisms (MLOs). They are small in size ranging from between 200 - 800nm.The cell of phytoplasmas contains both DNA and RNA.



(Fig.1) Electron microscopy picture of cross section of sieve tubes with phytoplasmas (6000×).

Transmission of phytoplasma

Transmission of phytoplasma is of two ways:

a) Movement between plants
b) Movement within plants

a) Movement between plants

- Phytoplasma are spread principally by insects of the families Cicadellidae, Delphacidae, Psyllidae, which feed on the phloem of the infected plants, ingesting phytoplasmas and transmitting them to the next plant on which they feed.
- Thus, the host range of phytoplasmas is strongly dependent upon the insect vector.

- Phytoplasma can overwinter in insect vectors or perennial plants.

- Phytoplasma enter the insect body through the stylet, pass through the intestine and then move to the haemolymph and colonize the salivary glands (The entire process can take up to 3 weeks).

- Once established in an insect host, phytoplasmas are found in most major organs.

- Phytoplasma can also be spread via dodders (Cuscuta) or by vegetative propagation such as the grafting of infected plant tissue onto a healthy plant.

b) **Movement within plants**

- Phytoplasmas move within phloem from a source to a sink, and can pass through sieve tube element. In this mode of transmission, Phytoplasmas spread more slowly.

Plant diseases caused by phytoplasma

- **Little leaf of brinjal**

  Vector: Leafhopper**(***Hishimonus phyticitis***)**

Symptoms:

- Leaves and plants become small.
- Nodes and internodes reduce in size.
- Yellowing of leaves.
- Bushy appearance.
- No fruiting if formed they become small and hard.

- **Sesame phyllody**

Vector: Leafhopper (*Orosius albicinctus*)

Symptoms:

- Plants are stunted
- Floral parts modified to leafy structures bearing no fruits and seeds.
- Virescence, proliferation, seed capsule cracking, formation of dark exudates on foliage and floral parts, and yellowing.

- **Rice yellow dwarf**

Vector: Leafhopper (*Nephotettix virescens*)

Symptoms:

- The infected plants are stunted
- Yellowish green to whitish green leaves.
- There is excessive tillering and leaves became soft and droop slightly.
- Root growth is also reduced significantly.

- **Kerala wilt disease of coconut**

Vector: Banana lace wingbug (*Stephanitis typica*), Planthopper (*Proutista moesta*)

Symptoms:

- Tapering of terminal portion of the trunk.
- Reduction of leaf size
- Abnormal bending or Ribbing of leaf lets termed as flaccidity.
- Flowering is delayed and also yield is considerably reduced.

- **Grassy shoot of sugarcane**

Vector: *Saccharosydne saccharivora, Matsumuratettix hiroglyphicus, Deltocephalus vulgaris and Yamatotettix flavovittatus*

Symptoms:

- **I**nitial symptom appears in the young crop of 3 - 4 months age as thin papery white young leaves at the top of the cane.
- Later, white or yellow tillers appear in large number below these leaves (profuse tillering).
- The cane becomes stunted with reduced internodal length with axillary bud sprouting.
- This disease appears in isolated clumps.

Other plant diseases caused by phytoplasma:

- Rice stripe
- Lethal yellowing of coconut
- Mulberry yellow dwarf
- Sandal spike
- Purple top wilt of potato
- Phyllody of marigold
- Peach X disease

References

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